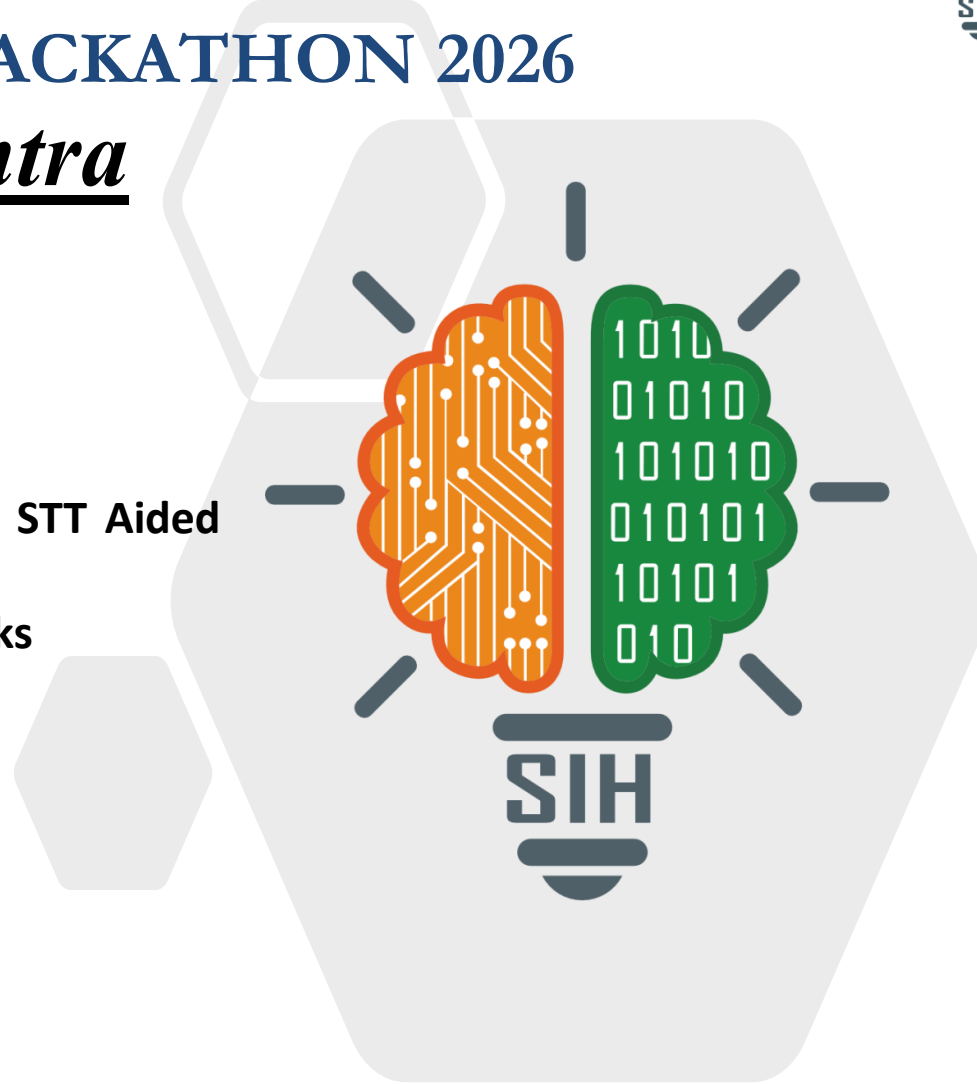


SMART INDIA HACKATHON 2026

iTantra

- Problem Statement ID – SIH26173
- Problem Statement Title- iTantra
- Problem Statement- Indian Multilingual TTS & STT Aided
Neural Transceiver Radio Access for low bitrate links
- Software Theme-Smart Automation
- PS Category- Software
- Team ID- 165027
- Team Name- GIT BIT



iTANTRA brings spoken communication to constrained links through local speech AI and compact text transport.

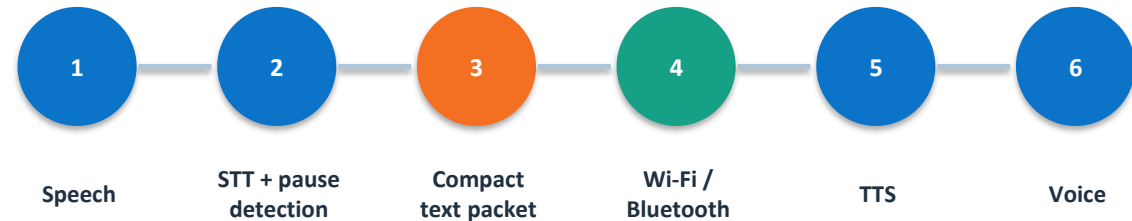


- Continuous voice is data-heavy
- Low-bitrate links cannot carry rich audio reliably
- Users still need spoken output for accessibility

ITANTRA APPROACH

Keep **VOICE** at both ends. Send only **COMPACT TEXT** in between.

On-device speech processing and direct Bluetooth / Wi-Fi Direct transport keep the core voice flow independent of cloud inference.



TEN LANGUAGES. ONE OFFLINE SPEECH PIPELINE.



Core design:

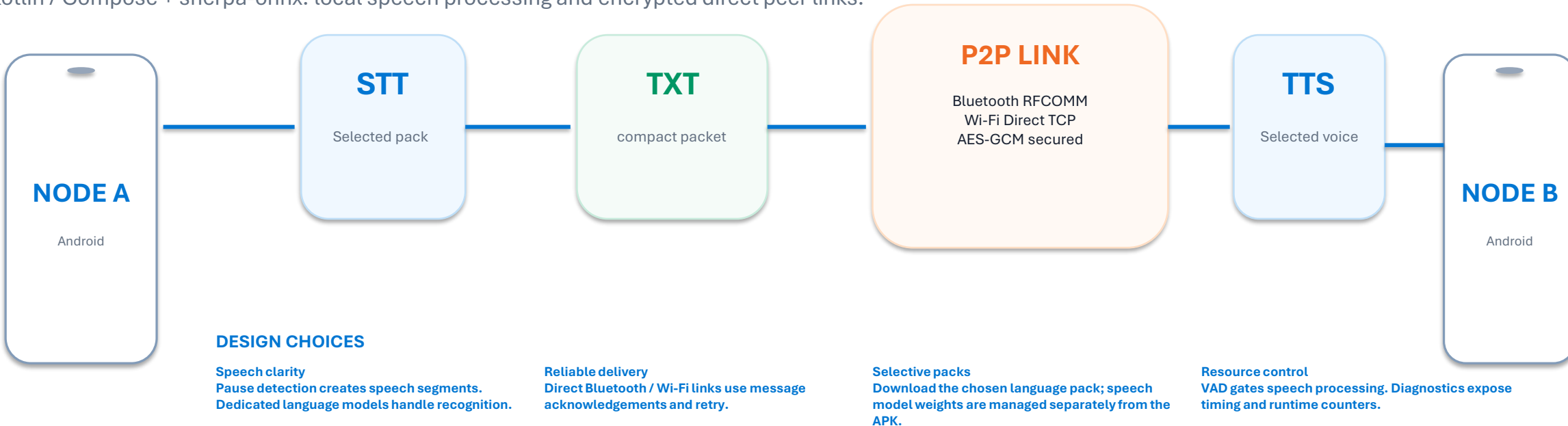
Speech -> local STT -> compact text -> local TTS

Selected speech packs run locally after initial model download.



TECHNICAL APPROACH & FEASIBILITY

Kotlin / Compose + sherpa-onnx: local speech processing and encrypted direct peer links.



APPLICATION FEATURES

- | | | | | | |
|-------------------|-----------------------------------|-------------------|-----------------------------------|-----------------|----------------------------------|
| ✓ Local speech | STT/TTS after model provisioning | ✓ Text transport | Compact messages instead of audio | ✓ Speech modes | PTT and pause-delimited segments |
| ✓ Emergency alert | Four compact SOS codes + playback | ✓ Encrypted peers | ECDH / SAS / AES-GCM session | ✓ Local history | Room / SQLCipher storage |

Demo flow: pair two Android devices -> push to talk -> encrypted text -> local voice playback

MEASURED PROGRESS & SUBMISSION PLAN



10 LANGUAGES → 454 TESTS → LOCAL AI

454 unit tests passed. Benchmarks cover ten languages; the Tamil GPU baseline provides the next training reference.

MEASURED RESULTS

Hindi WER 9.1%

English WER 6.9%

Tamil CTC WER 20.05%

Tamil RNNT WER 18.53%

Marathi TTS 181 ms

Tamil data overlap: 0

BENCHMARK CONTEXT

Hindi/English: desktop WER. Tamil: source GPU WER. 30 clips per set. Marathi: desktop synthesis-only mean.

Accuracy 40%

WER + speech quality

Efficiency 20%

APK / RAM / CPU

Latency 20%

Speech + delivery time

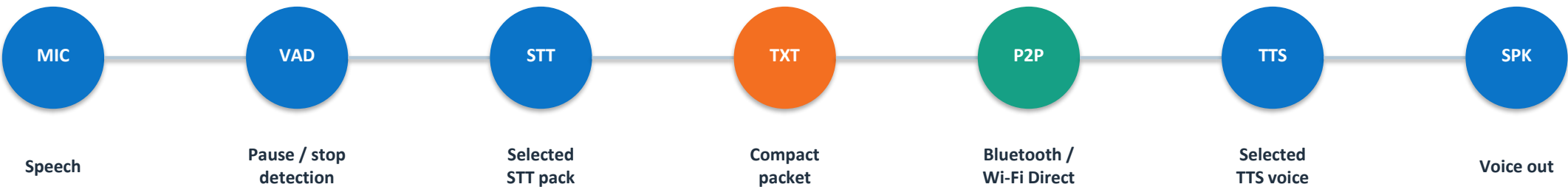
Demo plan

Two-device voice flow

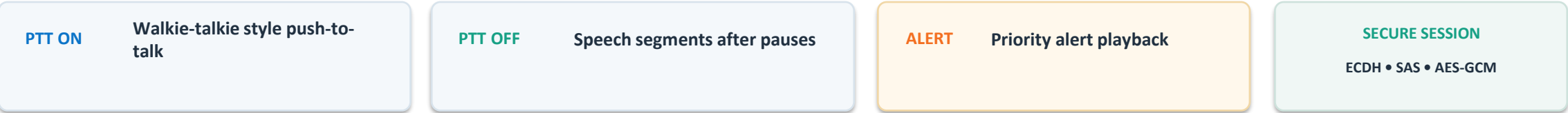
Git Bit

HOW iTANTRA WORKS — END TO END

CORE OFFLINE PIPELINE



OPERATING MODES



FINAL DEMO DELIVERY PLAN



SYSTEM ARCHITECTURE

REFERENCES & DELIVERY APPROACH

- Official problem SIH26173: sih.gov.in/sih2026PS
- Offline speech runtime: sherpa-onnx for Android
- STT: IndicConformer / English FastConformer
- TTS: MMS-VITS / Marathi Piper voice
- Speech benchmarks: FLEURS, 30 clips per language
- Tamil data: Kathbath, 4,428 train / 555 dev clips
- Validation design: speaker-separated splits + locked tests
- Delivery sequence: model tuning -> demo -> signed APK